

- |   |         |                                                                                                                                         |              |
|---|---------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1 | (a)     | Distance travelled before hitting the ground = Area under graph from 0 to 0.6 s.<br><br>$= \frac{1}{2} (4 + 10) * 0.60 = 4.2 \text{ m}$ | M1<br><br>A1 |
|   | (b)     | Distance that the ball bounce upwards<br>$= \frac{1}{2} (1.0 * 10.0) = 5.0 \text{ m}$                                                   | M1           |
|   |         | Height above original = 0.8 m                                                                                                           | A1           |
|   | (c)     | Acceleration = $10/1.0 = 10 \text{ m s}^{-2}$                                                                                           | A1           |
|   | (d)     | Time taken to travel upwards = time taken to travel downwards<br>Next time the ball will reach ground = 2.6 s                           | A1           |
| 2 | (a) (i) | conservation of linear momentum gives                                                                                                   | C1           |